

S403011 Machine Learning Project: Weather Prediction

1 Data Set and Goal

In Switzerland, talking about the weather is practically a national sport and for good reason: it changes fast! Your task is to use data from **10 weather stations** across the country to **predict the temperature in Bern**.

Weather forecasting in Switzerland is notoriously difficult, the Alps create countless microclimates, and even neighboring cities can experience completely different conditions. Accurately predicting the temperature in Bern is therefore more than a guessing game; it's a real data challenge.

In this project, you will use observations from **10 weather stations** across Switzerland to build machine learning models that **predict the temperature in Bern in 12 hours, in 24 hours and in 48 hours**. Reliable short-term forecasts are crucial for transportation, agriculture, and energy management.

We expect you to test a range of **machine learning methods** with varying levels of complexity and interpretability in order to identify the model that provides the most accurate temperature predictions. Beyond achieving good predictive performance, we encourage you to **gain meaningful insights into the weather patterns** underlying the data. This includes, for example, understanding how different meteorological variables influence temperature in Bern and how the spatial relationships among the 10 stations contribute to forecast accuracy.

1.1 Prediction Evaluation

For assessing the quality and the performance of your work, the metric we will use is the **mean absolute error**:

$$\text{MAE}(y, \hat{y}) = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

where n denotes the test size, and y_i and $\hat{y}_i$ denote respectively the true response and your prediction (for a given forecast time, e.g., 12 hours), for $i = 1, \dots, n$. Therefore, all the ML models you will use have to be adapted to this metric.

2 The Competition

Alongside the project, we will run a **data competition**, to test your model against your classmates and win a few bonus points (and bragging rights).

The competition is hosted on

<https://www.kaggle.com/competitions/s-403011-ml-unige-weather-prediction>

In order to join the competition, you will have to:

1. Create an account on <https://www.kaggle.com/>; please use a "Display name" that identifies yourself, for example your last name. When logged in, you can join the competition on

<https://www.kaggle.com/t/1b3787cc58b34360b9c6152b7650f992>

please do not share this link with anyone else outside of the course.

2. When all team members have joined the competition, you can create a team.

The competition ends on **19 December 2025 at 23h59**.

There are two different types of rankings on Kaggle:

1. **Public Leaderboard:** Each day you may submit up to two prediction files. These predictions are directly evaluated on 30% of the test data. This score is shown on the Public Leaderboard and gives you an indication of the accuracy of your prediction. The Public Leaderboard **does not count** for the final evaluation.
2. **Private Leaderboard:** Before the end of the Kaggle competition, you can choose one of your submissions for the final evaluation. After the end of the competition, the predictions of this submission are evaluated on the 70% remaining test data, resulting in the Private Leaderboard. The final scores in the Private Leaderboard determine the accuracy of your predictions and the competition winners. You should choose a final prediction that you believe performs best on new data, which might not necessarily be the best-scoring Public Leaderboard prediction.

3 Rules

1. You have to participate in teams of 3 students. Once your team is formed, a member must send an email, no later than **17 November 2025**, to christophe.valvason@unige.ch indicating the team members and your team's name, which must match the one on Kaggle.
2. Predictions should be based only on the training data and information from the Public Leaderboard.
3. You should use the Python programming language.
4. No cheating of any kind.
5. Use only ML models presented in lectures or seminars; you can use the various Python libraries presented during the seminars.
6. Refer to your course for every important step. You have to motivate your approach with explicit references to the course or the seminars (or e.g. the book "The Elements of Statistical Learning").
7. You have to explain and justify your main prediction approaches; please use the same notation as in the course.
8. Generative AI (ChatGPT, etc.) can be used as support; no complete text passages should be AI generated. Any AI use has to be documented; please follow the AI policy posted on moodle. Note that the project will be graded on your original contributions.

4 Prize

The first 3 teams in the Private Leaderboard receive bonus points for the course (which has 100 points in total).

- **1st place:** all team members receive 6 bonus points.
- **2nd place:** all team members receive 4 bonus points.
- **3rd place:** all team members receive 2 bonus points.

5 Deliverable and Grading

For this project, you have to submit two documents:

1. **The Python code:** your code has to be well documented and well structured. To evaluate your work, we expect to receive in return an *IPython Notebook*. It should contain your Python code corresponding to the Kaggle submission you selected for final evaluation. It should also output the same exact csv file uploaded on Kaggle, so make sure that your code is reproducible.
2. **A report:** you also have to submit a report explaining how you approached the problem and which methods you have used. Moreover, you should illustrate your findings with high quality and relevant plots. This report can be written in $\text{\LaTeX}$ or *Word*; $\text{\LaTeX}$ is a text editor which is widely used for writing technical reports or scientific papers; we also provide a $\text{\LaTeX}$ template. The final report should be 15 to 20 pages long.

The Python code and the report can be submitted on the Moodle course website where assignment modules will be created. Only one member per team has to submit the notebook, but make sure to mark the names of all team members and the Kaggle team name clearly on it. The report should describe in detail the data set and how you approached the problem. A possible structure might be:

1. **Introduction:** Description of the data set, main problem, challenges of ML and temporal data
2. **Exploratory Data Analysis:** Relevant information, structure of the data set, etc...
3. **Modeling:** description of the models, comparisons of methods, tuning parameters, model selection, etc...
4. **Final Result and Conclusion:** which model works the best for the problem, performance of the model, possible alternative and improvement, etc...

You can concentrate on the best model, but you should also compare the results (training, CV and test errors) with the other approaches. Grading will be based on scientific correctness, originality and presentation. The project will count for 30% **of your final grade**.